Type MS · Plot a line graph from a table of results

Answer 1 of 14 · Question 2 part a.i · 6 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number				
2	(a) (i)	• axes labelled with units; • axes correct way round and suitable scale; • points plotted correctly;; • straight lines joined; • lines identified;	time without with 0 65 110 30 120 215 60 130 230 90 95 200 120 80 190 150 65 180 180 65 170 don't include in published version	6
	(ii)	• without = 65 mg per 100cm^3; • with = 120 mg per 100cm^3;		2
	(iii)	any three from • without, level starts lower/level lower throughout; • smaller difference between highest and lowest values; • rises to lower levels/rise less quickly; • returns to normal (quicker)/same level 150-180;	ORA for person with diabetes	3
	(b)	• sample in a test/boiling tube; • add Benedict's reagent/solution; • heat in a water bath; • wear goggles; • brick red colour if glucose present; if candidate uses glucose testing sticks • sample in a test/boiling tube; • use of testing stick; • dip into urine; • compare colour with chart; • use of goggles;	Accept yellow/green/orange	5

Total 16 marks

Type MS · Plot a line graph from a table of results

Answer 1 of 14 · Question 2 part a.i · 6 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 2 of 14 · Question 5 part a · 5 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

5(a)	• correct labels on correct axes; • correct/appropriate scale; • correct plotting; • suitable lines; • lines labelled;	Allow line of best fit	**(5)**

Question Number	Answer	Additional guidance	Mark
5(b)	12.5 - 2.5; 10;	ECF from 5a	**(2)**

Question Number	Answer	Additional guidance	Mark
5(c)	Any three from: • exercise requires more energy; • from respiration; • more heat released raising body temperature; • more sweat released to cool body;	Reject energy produced/made Ignore anaerobic respiration	**(3 max)**

Total 10 Marks

Type MS · Plot a line graph from a table of results

Answer 2 of 14 · Question 5 part a · 5 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Paper 1: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Plot a line graph from a table of results

Answer 3 of 14 · Question 3 part b.i · 1 mark

2019 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

3	(a)	(i)	A (cornea and lens); B is incorrect as the iris does not play a role in focusing light C is incorrect as the iris does not play a role in focusing light D is incorrect as the pupil does not play a role in focusing light		1
	(b)	(i)	Points as follows: 150:2.7 200:3.2 250:2.3;		1
Clip (b)(i) with (b)(ii) and (b)(iii)		(ii)	Line of best fit missing out point at 150:2.7	Ignore lines that extended beyond points	1
		(iii)	150 lux/150:2.7/2.7 diameter;		1
			does not follow the pattern of other points/trend/not on the line of best fit;		1
		(iv)	As light intensity increases diameter decreases;	ORA	1
	(c)		Idea that distant/far away objects are blurred/difficult to see/ORA; as (light) rays do not meet on the retina/focused in front of retina/meet before the retina/light rays refracted too much;	Reject can't see	1 1

Total question 3 = 8 marks

Type MS · Plot a line graph from a table of results

Answer 3 of 14 · Question 3 part b.i · 1 mark

2019 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 4 of 14 · Question 5 part a.i · 6 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
5	(a)	(i)	• axes labelled (1) • suitable scale (1) • independent variable on X axis (1) • correct plots (2) • suitable curve (1)	-1 for each incorrect plot	6
		(ii)	• mean/average calculated (1) • reliable / reproducible / repeatable (1) • eliminate anomalies (1)		2
	(b)	(i)	70 x 100 (1) 90 78%/77.78/77.8 (1)	correct answer = full marks	2
		(ii)	• travels in the blood (1) • time to reach/affect nerve (1)	R heart	2
	(c)		• carries impulses to heart muscle (1) • reduces rate of contraction (1)		2

Total for question = 14 marks

Type MS · Plot a line graph from a table of results

Answer 4 of 14 · Question 5 part a.i · 6 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 5 of 14 · Question 4 part a.i · 5 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

4 (a)	(i)	• axes labelled (1) • suitable scale (1) • axes correct way round (1) • correct plots (1) • lines labelled/key given (1)		5
	(ii)	any two from • volume of water (1) • mass of carbohydrate/aspirin (1) • temperature (1) • time interval between each reading (1)	Allow type of aspirin Allow amount as an alternative to mass Ignore concentration	2
	(iii)	pH (1)		1
(b)		• (aspirin) dissolves/reduces pH more quickly (1) • (when combined) with sucrose (1) • so will act more quickly/relieves pain faster (1)		3

Total 11 marks

Type MS · Plot a line graph from a table of results

Answer 5 of 14 · Question 4 part a.i · 5 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 6 of 14 · Question 3 part b.i · 5 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

number				
3 (a)	(i)	A (amino acids); B and D are components of lipids C is a carbohydrate		1
	(ii)	• add Biuret solution to test solution; • mauve/lilac, protein present; • remains blue, protein absent;		3
(b)	(i)	• suitable axes labels; • suitable scale; • correct plots; • suitable curves; • curves labelled;	max 3 for bar chart accept key	5
	(ii)	• denaturation; • loss of shape of active site/enzyme; • substrate no longer fits/decreases rate of activity;		3
	(iii)	any three from • B is much more active/works faster; • over whole temperature range; • higher optimum temperature for B/works fastest at 80°C; • active over a greater range of temperatures;		3

Total 15 marks

Type MS · Plot a line graph from a table of results

Answer 6 of 14 · Question 3 part b.i · 5 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 7 of 14 · Question 2 part b.i · 3 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

2 (a) (i)		• a tube containing inhibitor and substrate; • but no enzyme;		2
	(ii)	any three from • concentration of enzyme; • concentration of inhibitor; • temperature; • pH; • same filter in colorimeter;		3
(b) (i)		axes labelled with units; axes correct way round; plots;		3
	(ii)	suitable curve;		1
	(iii)	similar shaped curve; reaching maximum sooner;		2
	(c)	same maximum never reached; despite increase in substrate concentration;		2

Total 13 marks

Type MS · Plot a line graph from a table of results

Answer 7 of 14 · Question 2 part b.i · 3 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 8 of 14 · Question 5 part b.i · 4 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

Question number					
5	(a)	(i)	$C_6H_{12}O_6$; $6H_2O$;		2
		(ii)	ATP breaks down/hydrolysed; to ADP and Pi;	Allow phosphate/P	2
	(b)	(i)	• suitable axis labels with units; • time on x axis; • suitable scale; • correct plots;		4
		(ii)	• suitable curve;		1
		(iii)	Five from; • mass of haemoglobin increases (with time/training at altitude); • increased number of red blood cells/more haemoglobin in the blood; • haemoglobin binds oxygen/red blood cells carry oxygen; • more oxygen transported around the body/in the blood; • to muscle (cells/tissue); • for aerobic respiration; • to release energy;		5

Total 14 marks

Type MS · Plot a line graph from a table of results

Answer 8 of 14 · Question 5 part b.i · 4 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 9 of 14 · Question 5 part b.i · 6 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b.i in this mark scheme extract

Question number					
5	(a)	(i)	any two from • to allow the heart rate to return to normal; • allow respiration/breathing rate/body temperature to return to normal; • to allow sweat to evaporate;		2
		(ii)	any three from • same time for exercise; • same type/intensity of exercise; • wear same clothing for each test; • measure sweat from same area;		3
	(b)	(i)	• axes labelled with units with independent variable on x-axis; • suitable scale; • both sets of data plotted correctly;; • suitable lines; • lines labelled;		6
		(ii)	any five from • exercise increases muscle activity; • more energy/ATP required; • increased respiration; • heat produced as by product; • lost from body; • by increased sweating; • increase in room temperature increases rate of sweating;		5

Total 16 marks

Type MS · Plot a line graph from a table of results

Answer 9 of 14 · Question 5 part b.i · 6 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 10 of 14 · Question 7 part a.i · 4 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number					
7	(a)	(i)	• axes labelled with units; • correct way round and suitable scale; • correct plots; • suitable line drawn;		4
		(ii)	• only not working correctly when rate falls below 100; • data/year not shown when that occurs/already below 100;		2
		(iii)	• 2021/2022; • rate falling at 5-10 units per annum/similar values each year;		2
		(iv)	• not very accurate/results vary/fluctuate; • earlier data showed improvement in function; • could happen again/results may change;		3
	(b)		any three from • protein broken down into urea/deamination; • toxic; • needs to be excreted; • incorrectly functioning kidney can't remove it all if there is a lot; • less protein less urea;		3

Total 14

Type MS · Plot a line graph from a table of results

Answer 10 of 14 · Question 7 part a.i · 4 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Plot a line graph from a table of results

Answer 11 of 14 · Question 4 part a.i · 3 marks

2024 JUN · Paper 01 (regional R) · find part a.i in this mark scheme extract

Question number				
4	(a) (i)	• axes labels and units and right way around; • linear scale and at least half grid; • correctly plotted points to half square tolerance;		(3)
	(ii)	• curve of best fit;	do not accept straight line going through all points	(1)
	(iii)	6		(1)
	(iv)	• time taken to digest starch decreases until pH 6; • time taken to digest starch increases (from/after PH6) to pH10;	Allow 1 mark for decrease and then increase in time as pH increases	(2)
	(v)	• reference to pH2 not being optimum/best pH for **enzyme** (activity)/pH6 being (closer to) optimum pH for enzyme; • enzymes denatured / active site changes shape/deformed (at pH2); • less/no binding to substrate/ substrate cannot fit into active site/ fewer enzyme -substrate complex;		(3)
	(vi)	small intestine/ mouth/duodenum;		(1)
	(b)	• add iodine solution; • blue-black/blue/black colour indicates the presence of starch;		(2)

Total for Question 4 = 13 marks

Type MS · Plot a line graph from a table of results

Answer 12 of 14 · Question 2 part a.i · 3 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
2	(a)	(i)	• axes labels; • plots; • scale;	axes correct way round use aty least half of the graph paper for the plotted results ignore extrapolation of line	3
		(ii)	line of best fit;		1
		(iii)	As time increases pH decreases;	ECF from an incorrectly drawn graph	1
	(b)		Any three from: • lipase; • digests/breaks down fat (in milk); • to fatty acids; • which lowers pH/increases acidity;		3
	(c)		time;	allow mins/minutes	1
	(d)		Any two from: • temperature; • volume/concentration of enzyme/substrate (solution)/volume of milk; • type of milk;	ignore amount	2
	(e)		repeat the experiment		1

Total Marks 12

Type MS · Plot a line graph from a table of results

Answer 12 of 14 · Question 2 part a.i · 3 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 13 of 14 · Question 4 part a.i · 4 marks

2025 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

4	(a)	(i)	• linear scale/plotted on half of grid; • correct way round; • axes labelled with units; • all points plotted correctly;	mp4- half-square tolerance	4
		(ii)	draw curve of best fit;	Ignore beyond the 2 extremes when line is drawn. 2 straight lines=0 Slight curve at 4 is enough for 1 mark	1
	(b)	(i)	• pepsin digests/breaks down protein; • (to)amino acid/peptides; • (which is)soluble/dissolves; • black silver salts removed /fall away/no longer present;	Mp1- accept hydrolyse MP4- accept silver salts broken down	3
		(ii)	• 1//2/1 to 2; • optimum pH for pepsin;		2
		(iii)	• temperature; • area of film/size/length/width/thickness/; • volume of (pepsin) solution; • type of photographic film/amount of protein in film; • pH;	Mp3-ignore amount	3
	(c)		• adding more enzyme/pepsin has no effect on reaction/ maximum rate of reaction at 4%; • all protein/substrate digested/ substrate is a limiting factor/not enough protein to digest; • enzyme no longer a limiting factor/ active sites empty;	MP1- accept Vmax not enough/no more substrate/protein to bind to active sites= 2 marks	2
					Total = 15

Type MS · Plot a line graph from a table of results

Answer 13 of 14 · Question 4 part a.i · 4 marks

2025 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plot a line graph from a table of results

Answer 14 of 14 · Question 4 part c.i · 5 marks

2025 JUN · Paper 01 · extract 1 of 2 · find part c.i in this mark scheme extract

4 (a)		disease causing organism/micro-organism;		1
(b)	(i)	any three from • cytoplasm; • cell/plasma membrane; • cell wall; • (loop of) nucleic acid/DNA; • flagellum;	if more than three parts labelled minus one mark for each incorrect label	3
	(ii)	• distance between A and B = 64 mm/6.4 cm; • 64 × 1000 = 64 000 µm; • magnification = $\frac{64\,000}{2}$ = 32 000;	ecf allow +/- 1 mm	3
(c)	(i)	• suitable scale, axes labelled with units; • correct way round; • points plotted correctly; • lines joining plotted points; • key to identify lines;	needs to cover half of paper award mp1,2 and 5 if bar chart plotted	5
	(ii)	any three from • both fall/vaccine becomes less effective over time; • 45-64 years more effective (over all months)/over 65 less effective (over all months); • slowly for first two months; • more/ rapid /steep decrease, between 3-4 months/after month 3 (for over 65);		3
				15 marks

Type MS · Plot a line graph from a table of results

Answer 14 of 14 · Question 4 part c.i · 5 marks

2025 JUN · Paper 01 · extract 2 of 2 · find part c.i in this mark scheme extract

Question number	Answer	Notes	Marks